Wavvar MMWK Module Documentation

MINI Module Introduction

Wavvar Technologies

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1. Module Overview

`MINI (RPM)` is part of the `RPX 6843` sensing module family. The `6843` series is Wavvar's flagship module product line, built on TI's high-performance millimeter-wave radar technology for advanced spatial sensing scenarios that require stable motion tracking and accurate spatial measurement.

Typical applications include:

- Human presence detection and fall detection
- Dynamic trajectory tracking
- Occupancy detection
- Activity recognition
- Entry and exit monitoring
- In-bed / out-of-bed detection
- Point cloud data visualization

2. Appearance and Dimensions

MINI appearance reference

Item	Specification
Dimensions	65x65x18 mm

3. Technical Specifications and Key Features

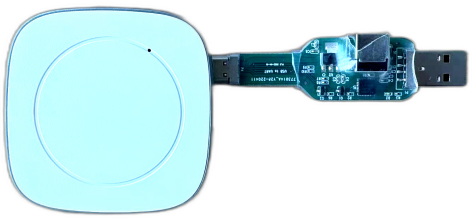
Category	Item	Specification
Power	External Supply	5V DC 2A
	Adapter	100-240V AC input
	System Power Consumption	< 10W
Operating Parameters	Installation Method	Ceiling mount or wall mount
	Maximum Detection Range (Wall-mounted)	Depends on installation height, tilt angle, target reflection characteristics, and algorithm configuration; configurable
	Field of View (FOV)	Approximately 120°-140° horizontally (estimated from the antenna pattern; actual coverage depends on installation method, enclosure structure, and algorithm configuration)
	Operating Temperature	0°C to 45°C (system ambient temperature)
	Operating Humidity	< 95% (non-condensing)
	Wall-mount Pitch Angle	30° (downward tilt)
Radar Characteristics	RF Frequency Band	60-64 GHz
	Tx/Rx Channels	3TX / 4RX (see Note 1)
	Modulation	FMCW
	Output Power per TX Channel (EIRP)	15 dBm
Connectivity and Integration	Cloud Protocols	MQTT, HTTP, HTTPS
	Wi-Fi	Wi-Fi 802.11 b/g/n (2.4 GHz)
		Station / SoftAP / Station + SoftAP
		Up to 150 Mbps (theoretical; actual performance depends on the network environment)
	Local Communication	UART (data format is defined by firmware and can support binary or JSON)
Hardware Architecture	Processing Architecture	Dual-chip heterogeneous architecture (mmWave radar SoC + main MCU)
	Radar Processing Unit	ARM Cortex-R4F + C674x DSP + Hardware Accelerator (HWA)
	Main Controller	ESP32 dual-core processor
	On-chip Memory	520 KB (ESP32) + 1.75 MB (radar SoC)

Category	Item	Specification
	PSRAM	8 MB PSRAM (connected to ESP32)
	Flash Storage	8 MB (ESP32) + 4 MB (radar SoC, optional)
	I/O and Indicators	1x LED, 1x key, 1x LED (optional)
	IMU (Optional)	Optional 6-axis gyroscope + 3-axis accelerometer
	Ambient Light Sensor (Optional)	Optional support

Note 1: For the “Tx/Rx Channels” parameter, up to 2TX simultaneous transmission is supported in 1.3 V mode; 3TX simultaneous transmission requires the 1V LDO bypass operating mode.

4. Development and Flashing Orientation

To ensure successful flashing and serial communication, pay attention to the required `Type-C` orientation. When connecting the flashing debugger to a `Mini` device, align the `A` side of the debugger with the front side of the enclosure.

Platform	Alignment Diagram
Mini Device	

5. Interface Description

The `MINI` module interface description includes the flashing debugger `Type-C`, the status `LED`, and the key input.

5.1 Type-C Interface on the Flashing Debugger

The `USB-to-UART V1.3` debugger board is used for firmware flashing and serial console access. Both debugger variants use the same `Type-C` pin definition and communication capability.

If the port is used only for power delivery, the module does not distinguish between side `A` and side `B`. When used for communication, the module `Type-C` port is side-sensitive. The pin definition is as follows.

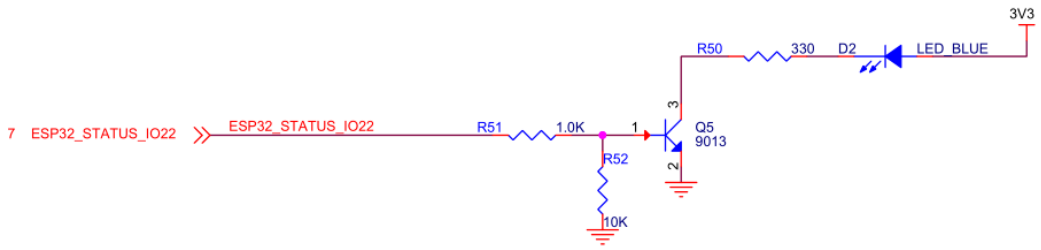
Type-C	Definition
A5	UART_RX
A6	RTS
A7	DTR
B8	UART_TX
A1/A12/B1/B12	GND
A4/A9/B4/B9	5V input

The `USB-to-UART V1.3` flashing debugger pin definition is as follows.

Pin	Color	Signal
A5	Orange	RX
A6	Green	RTS
A7	Blue	DTR
B8	Yellow	TX
GND	Black	GND
VBUS	Red	5V

5.2 Status LED Interface Reference

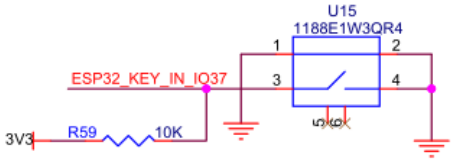
The status `LED` is controlled by `ESP32_STATUS_IO22`. The circuit reference is shown below.



MINI status LED interface reference

5.3 Key Interface Reference

The key input is connected to `ESP32_KEY_IN_IO37` and is pulled up to 3V3 through a `10kΩ` resistor by default. Pressing the key triggers a low-level input.



MINI key interface reference

6. Related Documents

1 RPX Series Usage Guide